

Module Firmware

Each module has one or two microcontrollers with firmware that manages their module's set of inputs and outputs. Typically, these are specialized based on Control (CTRL) or Display (DSPL), meaning buttons/switches or LED outputs respectively. The only alternative is the Telemetry module which runs an Arduino Nano that handles both as well as the screen drawing functionality.

Both types of firmware utilize the i2c protocol and use the TinyWire library.

CTRL

The CTRL firmware uses the ShiftIn library to collect inputs, as bits, which it stores in bytes. When requested by the master (Wire.requestFrom(ctrl_address, in_bytes);), it provides one or two bytes.

```
void onI2CRequest() {
    TinyWire.send(bytes[0]);
    TinyWire.send(bytes[1]); //Optional
}
```

DSPL

The DSPL firmware uses the ShiftRegister74HC595 library to drive LED arrays. When it receives data from the master in the form of bytes, it converts them into serial bits.

```
void onI2CReceive(int howMany) {
    bytes[0] = TinyWire.read();
    bytes[1] = TinyWire.read(); //Optional
}
```

Analog

In the Analog firmware, digital inputs are collected and condensed into bytes in the same manner as CTRL firmware. Each joystick axis is read using the microcontroller's ADC, producing a 10-bit value. Because this value cannot fit into a single byte, each axis is transmitted as two bytes containing the high and low portions of the ADC value. It also sends a byte of other input data.

```
axis1 = analogRead(A1);
axis2 = analogRead(A2);
axis3 = analogRead(A3);

void onI2CRequest() {
    TinyWire.send(bytes[0]); // Other inputs
    TinyWire.send(highByte(axis1));
    TinyWire.send(lowByte(axis1));
    TinyWire.send(highByte(axis2));
    TinyWire.send(lowByte(axis2));
    TinyWire.send(highByte(axis3));
    TinyWire.send(lowByte(axis3));
}
```
